

GIS Training Courses

Indian Institute of Remote Sensing (IIRS), Dehra Dun

Post graduate Diploma courses and short-term courses in GIS and Remote Sensing
www.isro.org/iirs-training.htm

National Remote Sensing Agency (NRSA), Hyderabad

Courses on Remote Sensing technology and applications
www.nrsta.gov.in

Anna University, Chennai

Post graduation in Remote Sensing
www.annauniv.edu
www.engr.uky.edu/~arunp/anna/irs.html

Jawaharlal Nehru University (JNTU), Hyderabad

Post graduate degree in Spatial Information Technology and Doctorate in Remote sensing
www.jntu.ac.in/crs.html

GIS Institute, NOIDA

Short-term courses in GIS software and applications. It is run by the Centre for Spatial Database Management and Solution
www.gis institute.net

ESRI India

Short-term courses in GIS software and database design
www.esriindia.com

Pentasoftware Education

Short-term courses in GIS software and database integration
www.pentasoftware.com

Note: This is a listing, not a recommendation.

architect is to develop, design and maintain spatial databases. If you plan to step into the shoes of a data architect, familiarity with spatial data models and their architecture is essential. "Since Oracle is a popularly used database, proficiency in Oracle is sufficient and knowledge in the latest version of Oracle Spatial is advantageous," says Yogendran. Robins John, a GIS professional with Speck Systems, says that attribute data linked with spatial data makes 'intelligent data', which aids decision-makers in



An analytical mind and the ability to extract information from maps are essential. A background in mathematics would help you with the mathematical analysis in GIS projects

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querying and analysing information. The attribute data related to the spatial data may be stored in any RDBMS such as Oracle, MS Access or MySQL.

Analyse this!

Data analysis in GIS uses the spatial and non-spatial information in a database to answer queries and provide solutions. Data analysis is done in the following areas: database query (selectively retrieving and displaying information from a database), overlaying of maps (vector-to-raster and raster-to-vector conversions), network analysis, surface modelling, geo-calculations, geo-statistics and others.

GIS analysts or professionals are the ones who carry out data analysis in a project. "A key requirement for a GIS analyst is domain expertise and the ability to visualise GIS-based structured solutions," says Prasad. "A crop monitoring system needs an expert in agriculture with a doctorate in one of its sub-domains and good knowledge of image processing and GIS," he continues. Strong analytical skills and versatility to adapt to project requirements on an industry-to-industry basis have to be the strong points of a GIS analyst. "Formal qualification definitely comes into the picture with GIS data analysis. This needs specific educational qualifications and experience in the related field," says John. According to him, for analysis, a degree or post graduation in Geology, Geography, Environmental Science, Remote Sensing, Geo-Information Technology, Town planning or Civil

Engineering is desirable, supplemented with knowledge of CAD and GIS software.

Yogendran's job profile as a GIS analyst in Marine GIS involves project planning, execution, management and client interaction. He works on digitisation, data capturing and creation of nautical charts using a GIS software called CARIS as well as compatibility of charts with standards adopted by the International Hydrographic Organization. "GIS application development and GIS analysis are the hot and happening areas and promise growth," he states.

Is there a business opportunity?

If GIS fascinates you but you don't want to be involved with the technical aspects, there's another avenue open—business development. "Opportunities exist for marketing GIS software and 'concept selling' of the technology and its applications," says John. The qualifications required would be an MBA with sufficient knowledge

of the GIS field. John adds, "The career advances are like that of any marketing executive." He started out in the data conversion field but is Resident-in-charge of the Botswana office of Speck Systems, a GIS company based in Hyderabad.

What must I do to go the GIS way?

Sufficiently interested to consider GIS as a career option? You can now avail of formal education as

Tools of the Trade

The software and tools used in a GIS system range from creating the front end map to GIS software used for data analysis and visualisation. For the front end graphical output or the maps, CAD software such as Microstation and Auto CAD Map is used. The most popular GIS packages are ArcInfo and ArcView GIS from vendors like ESRI, Geomedia from Intergraph, CARIS GIS, SICAD, MapInfo, MapObjects and ArcSDE. The choice of software depends on the industry applications. For instance, CARIS is used for marine GIS and electronic navigation systems.

ERDAS is the software used for processing satellite remote sensing images and aerial photographs.

GIS data can also be Internet enabled. For this the software used are Auto CAD Map Guide and MapXtreme.